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Biology Mock

Answers and Marking Scheme

B: independent mark (1 mark each) M: method A: answer (with unit) C: communication

Underlined words are key words. "/" shows an acceptable alternative; for "any n", only the first n points are marked; contradictory answers score no marks. For revision only; other valid answers should also be credited.

Section A Human Physiology: Regulation and Control

1. (20 marks)

(a)(i) (2)

B1 FSH rises from 7 to 85 IU L^{-1} (about 12 times), especially fast after 48 (1)

B1 oestrogen falls from 140 to 18 pg mL^{-1} (1)

〔max. 2〕

Also accept：以「促卵泡激素上升超過 10 倍」或「雌激素下降至原來約八分之一」描述，有數據支持即可

Do not accept：只寫「上升」「下降」而沒有數值

Common error：只寫「上升」「下降」而沒有引用數值。

(a)(ii) (4)

B1 very few follicles remain in the ovaries, so the ovaries respond poorly to FSH (1)

B1 fewer follicles develop, so less oestrogen is secreted (1)

B1 normally oestrogen inhibits the secretion of FSH by the pituitary gland (negative feedback) (1)

B1 with low oestrogen, the inhibition is removed and the pituitary secretes more FSH (1)

B1 even more FSH cannot stimulate the ovaries to secrete enough oestrogen, so FSH keeps rising (1)

〔any 4；max. 4〕

Also accept：「負反饋失效／負反饋減弱」可代替第三及第四點中的「負反饋」字眼

Common error：只寫「負反饋」而沒有說明雌激素作用於垂體，或沒有指出卵巢對促卵泡激素沒有反應。

(a)(iii) (4)

B1 benefit: fewer fractures with HRT (16 to 10 per 1000) (1)

B1 benefit: women with marked relief of hot flushes rise from 25% to 80% (1)

B1 risk: breast cancer rises from 12 to 15 and blood clots from 5 to 9 per 1000 (1)

B1 the absolute differences per 1000 women are small (only 3–6) (1)

B1 conclusion: benefits and risks differ between individuals (e.g. family history, risk of osteoporosis, severity of symptoms), so each woman should decide with her doctor; "all women should" is not justified (1)

〔any 4；max. 4〕

Also accept：以「骨折減少約 40%、乳癌增加 25%」等相對變化表達亦可；指出數據只覆蓋 5 年，較長期的影響未知（可代替第四點）

Common error：只列出好處或只列出風險，沒有作出平衡的結論。

(b)(i) (4)

B1 cold receptors in the skin detect the fall in temperature, and the hypothalamus causes the skin arterioles to constrict (vasoconstriction) (1)

B1 much less warm blood flows to the skin, so skin temperature falls rapidly to near water temperature (1)

B1 much less heat is carried from the core to the body surface, so heat is retained in the core (1)

B1 the skin and subcutaneous fat act as an insulating layer, reducing heat transfer from the core to the water (1)

B1 the temperature difference between skin and water becomes smaller, lowering the rate of heat loss (1)

〔any 4；max. 4〕

Also accept：「皮膚溫度大跌表示皮膚血流量減少」可代替第二點

Do not accept：血管移離皮膚表面；皮膚「關閉」毛孔

Common error：寫成「血管向下移」；或只描述數據，沒有解釋熱如何留在體心。

(b)(ii) (3)

B1 metabolic rate rises rapidly from 1 (rest) to about 3.2–3.4 and then stays steady (1)

B1 skeletal muscles shiver, i.e. contract rapidly and involuntarily (1)

B1 respiration in the muscles increases, producing more heat (1)

〔max. 3〕

Also accept：接受「升至約 3 倍」；「腎上腺素／甲狀腺素令細胞代謝加快」可代替第二點

Do not accept：因游泳運動令代謝率上升（題目已註明身體不游動）

Common error：把代謝率上升歸因於運動，或沒有說明顫抖產熱。

(b)(iii) (3)

B1 after leaving the water, the skin warms up and its blood vessels dilate, so more blood flows to the skin and limbs (1)

B1 blood cooled in the limbs and skin returns to the core, lowering the core temperature (1)

B1 heat flows from the core to the still-cold peripheral tissues (1)

B1 evaporation of seawater from the skin removes heat (1)

B1 metabolic rate falls gradually, so less heat is produced (1)

〔any 3；max. 3〕

Also accept：「岸上 20°C 仍低於體溫，身體繼續散熱」可代替第四點

Common error：以為上岸後體溫會立即回升，沒有用圖中皮膚溫度回升作提示。

Section B Applied Ecology

2. (20 marks)

(a)(i) (2)

B1 species P: mercury content is very low (about 0.03–0.08 μg g^{-1}) and rises only slightly with body length (1)

B1 species Q: similar to P below about 30 cm; above that, mercury content rises sharply with length, reaching about 0.9 μg g^{-1} (1)

〔max. 2〕

Also accept：以「Q 體長約 65 cm 以上超過 0.5 μg g^{-1} 指引值」描述 Q 的急升亦可

Common error：只描述一個物種，或沒有引用數值。

(a)(ii) (4)

B1 Q below about 30 cm has a mercury content similar to P, so young Q probably feeds on plankton like P (1)

B1 mercury rises sharply above about 30 cm, so Q probably switches to eating other fish, at a higher trophic level (1)

B1 each predatory fish eats many prey; mercury is not broken down or excreted and builds up, so its concentration exceeds that in the prey (1)

B1 larger fish are older and have been accumulating mercury for longer (1)

〔max. 4〕

Also accept：「Q 長大後轉食較大的獵物，營養級上升」可代替第二點；使用「生物放大」「生物積累」等用語而意思正確亦給分

Common error：只寫「*Q* 是捕食者所以汞較高」，沒有用 *30 cm* 以下與 *P* 相近的數據推論食性改變。

(a)(iii) (4)

B1 support: species P and small species Q have mercury contents far below the guideline value of 0.5 μg g^{-1} (1)

B1 mercury can damage the development of the fetal nervous system, so pregnant women should limit intake (1)

B1 fish provides protein and beneficial fatty acids, so limiting rather than banning fish is reasonable (1)

B1 limitation: data come from only one water body and two species; serving size is not stated (1)

B1 limitation: consumers buying fish fillets or pieces cannot tell the species or size of the fish (1)

〔any 4；max. 4〕

Also accept：胎兒對汞比成人敏感，所以孕婦的指引應較一般人嚴格；物種 Q 體長約 60 cm 以下仍低於指引值，「體型較小」的界線不清楚

Common error：只複述建議，沒有引用數據或提出不足之處。

(b)(i) (2)

B1 to show whether these parameters would change between 2020 and 2025 without the reef (1)

B1 the control site hardly changed, so the changes at the reef site can be attributed to the reef rather than other factors such as weather or sewage discharge (1)

〔max. 2〕

Also accept：「排除年份之間的自然變化」可代替第一點

(b)(ii) (4)

B1 oysters filter-feed, removing phytoplankton and suspended particles, so chlorophyll a falls from 22 to 9 and suspended solids from 31 to 14 (1)

B1 fewer dead phytoplankton sink, so less oxygen is used by aerobic decomposers on the seabed (1)

B1 clearer water lets light reach the seabed, where algae / seagrass photosynthesise and release oxygen, so dissolved oxygen rises from 3.8 to 5.9 (1)

B1 the three-dimensional reef provides habitats and shelter, and with more dissolved oxygen, benthic species increase from 11 to 26 (1)

〔any 4；max. 4〕

Also accept：蠔殼表面提供附著面供其他生物生長（可代替第四點的「立體結構」）

Common error：只描述數據變化，沒有連繫到蠔的濾食作用。

(b)(iii) (4)

B1 support: oysters can reduce phytoplankton and suspended solids, improving water quality (1)

B1 limitation: oysters cannot remove dissolved pollutants (e.g. ammonia, phosphate) or pathogens (1)

B1 limitation: the volume of urban sewage is huge and reefs filter only a limited amount; reefs take years to establish, and low oxygen or storms can kill many oysters (1)

B1 limitation: oysters accumulate heavy metals and pathogens and become unsafe to eat (1)

B1 conclusion: oyster reefs can only supplement, not replace, sewage treatment (1)

〔any 4；max. 4〕

Also accept：污水處理廠可消毒、降低生化需氧量及除去氮，蠔礁不能做到（可代替第二點）

Common error：只同意或反對而沒有理據，或忽略蠔不能處理溶解的污染物。

Section C　Microorganisms and Humans

3.　(20 marks)

(a)(i) (2)

M1　10^(5.9 − 1.0) = 10^4.9 (1)

A1　≈ 7.9 × 10^4 times (accept 7.9–8.0 × 10^4) (1)

〔max. 2〕

Also accept：答「約 80 000 倍」亦可

Do not accept：4.9 倍

Common error：直接把對數相減，答 *4.9* 倍。

(a)(ii) (4)

B1　germination needs warm, moist conditions, which are also ideal for bacterial growth (1)

B1　germinating seeds release nutrients such as sugars that support bacterial growth (1)

B1　bacteria hide in cracks on the seed surface and are hard to remove by washing (1)

B1　even a small initial number can multiply to an infective dose within a few days (e.g. to 10^5.9 at 25°C) (1)

B1　sprouts are often eaten raw or only lightly cooked, so bacteria are not killed (1)

〔any 4；max. 4〕

Also accept：「芽菜在 35°C 萌發時菌數最高 (10^7.3)」可作為溫暖條件的數據支持

Common error：只說「芽菜不潔」，沒有連繫萌發條件與細菌生長。

(a)(iii) (4)

B1　Expt 1: hot water reduces bacteria from 10^5.0 to 10^2.0 (a fall of 3.0), more than chlorine (to 10^3.2, a fall of 1.8) (1)

B1　Expt 2: hot-water-treated sprouts have the fewest bacteria (10^2.6) but germination falls from 96% to 84%; chlorine gives 93% germination but 10^3.9 bacteria (1)

B1　neither method kills all bacteria; survivors multiply again during germination (1)

B1　conclusion: hot water is safer, and food safety matters more than a small fall in germination (or the reverse, with a reason) (1)

B1　consumers: cook sprouts thoroughly / store at 4°C or below and eat them soon (1)

〔any 4；max. 4〕

Also accept：「兩種方法合併使用」或「處理後以冷水沖洗、低溫萌發」作建議亦可

Common error：只選出熱水處理而沒有考慮萌發率，或沒有指出兩種方法都不能除盡細菌。

(b)(i) (2)

B1　pH falls from 6.2 to 4.2 (1)

B1　lactic acid bacteria increase from 10^4.0 to 10^8.8 per g, about 6 × 10^4 times (1)

〔max. 2〕

Also accept：「乳酸菌在第 0–2 日增加最快」可配合數據描述

Common error：沒有留意縱軸是對數尺度，把 *4* 至 *8.8* 當作只增加約兩倍。

(b)(ii) (4)

B1　lactic acid bacteria increase sharply on days 0–2 (from 10^4.0 to 10^8.2) (1)

B1　lactic acid bacteria ferment sugars, producing much lactic acid (1)

B1　pH falls from about 5.9 on day 1 to about 4.2 on day 4 (1)

B1　the acidic conditions denature the enzymes of other bacteria / stop their growth, so other bacteria decline after day 1 (1)

B1　the huge number of lactic acid bacteria compete with other bacteria for nutrients (1)

〔any 4；max. 4〕

Also accept：鹽令細胞以滲透作用失水、密封容器缺氧，都可抑制其他細菌（可作補充，但這些條件第 0 日已存在，不能單獨解釋第 1 日後的下降）

Common error：只以鹽或缺氧解釋，未能說明為何第 *1* 日後才下降。

(b)(iii) (4)

B1 (1) the juice contains many lactic acid bacteria, so fermentation starts with many of them and pH falls faster (1)

B1 (1) the early rise of other bacteria (from 10^5.0 to 10^5.8 on day 1) is reduced, improving safety (1)

B1 (1) but old juice may also bring in yeasts, moulds or other contaminants (1)

B1 (2) with less salt, other bacteria are less inhibited and may multiply in the first day or two before pH falls, increasing the risk of spoilage or food poisoning (1)

B1 (2) combining it with suggestion (1) to lower pH quickly may partly offset the risk of reducing salt (1)

〔any 4；max. 4〕

Also accept：(2) 減鹽亦可能令乳酸菌生長得更快，pH 更早下降（須配合理由）

Common error：只評價一項建議，或沒有引用圖中其他細菌數目的變化。

Section D Biotechnology

4. (20 marks)

(a)(i) (3)

B1 inject HCG (antigen) into a mouse to stimulate B lymphocytes producing anti-HCG antibodies (1)

B1 remove B lymphocytes and fuse them with myeloma (cancer) cells to form hybridoma cells that divide continuously (1)

B1 select the hybridoma producing only anti-HCG antibody, culture it in large numbers and collect the antibodies (1)

〔max. 3〕

Also accept：「B 細胞」「淋巴細胞」可代替「B 淋巴細胞」

Common error：只寫「把 *B* 細胞與癌細胞結合」，沒有說明篩選產生特定抗體的細胞。

(a)(ii) (3)

B1 HCG in the urine binds to the red-labelled antibodies in zone R, forming complexes that move with the liquid (1)

B1 the antibody at T also binds to HCG, at a different site from the antibody in zone R (1)

B1 the complexes are held by the antibodies at T, so red particles accumulate there (1)

B1 without HCG, the red-labelled antibodies cannot be captured at T and flow past it (1)

〔any 3；max. 3〕

Also accept：「T 線的抗體是抗 HCG 抗體」可代替第二點的前半部分

Common error：以為 *T* 線的抗體直接與區域 *R* 的抗體結合（那是 *C* 線）。

(a)(iii) (2)

B1 W: pregnant (positive); X: not pregnant (negative) (1)

B1 Y and Z: no line C, so the results are invalid and must be retested (1)

〔max. 2〕

Also accept：「C 線顯示液體已流過 T 線、抗體仍有作用」可作為判斷 Y、Z 無效的理由

Common error：把 *Y* 讀作懷孕，忽略 *C* 線沒有出現。

(a)(iv) (2)

B1 before about 13 days after fertilisation, HCG in urine is below the minimum detectable level of 25 mIU mL^{-1}, so line T does not appear even if pregnant (1)

B1 test on or after about day 14 after fertilisation (the expected day of menstruation) (1)

〔max. 2〕

Also accept：接受「第 13–15 日之後」；「若結果陰性，數日後再驗」可作為補充

Common error：只寫「*HCG* 太少」而沒有引用最低檢測值或日數。

(b)(i) (3)

M1 $18 \times 12 = 216$ mg; $216 \div 4$ (1)

A1 = 54 glands (1)

B1 the number of donated glands is limited, so few children can be treated / supply is short and cost high (1)

〔max. 3〕

Also accept：若產品純度只有 70%，實際需要更多垂體（約 77 個），亦可給第三分

Common error：忘記乘以 *12* 個月。

(b)(ii) (3)

B1 bacteria reproduce rapidly and can be grown on a large scale in fermenters, giving a large yield (about 1 g per 10 dm^3) (1)

B1 the product does not come from human tissue, so there is no risk of transmitting disease agents from donor brains (1)

B1 the bacteria carry the human growth hormone gene, so the product is identical to the human hormone and over 99% pure (1)

B1 supply is reliable and cost is lower (1)

〔any 3；max. 3〕

Also accept：「不須依賴捐贈者」可代替第四點

(b)(iii) (4)

B1 for: extra height may improve the child's self-esteem and social life (1)

B1 against: the effect is small (about 4 cm on average) but requires daily injections for five years, causing pain and inconvenience (1)

B1 against: long-term side effects are not fully known; it is expensive and may take resources from patients who truly lack the hormone (1)

B1 ethics: short stature is not a disease; treating normal variation as illness may reinforce social bias about height; children may not be able to give informed consent (1)

B1 a reasoned conclusion (1)

〔any 4；max. 4〕

Also accept：「4 cm 只是平均值，個別兒童的效果可能更大或更小」可代替第二點的分析

Common error：只從單一角度（例如只談費用）作答，沒有權衡利弊。